

Topic 10 :

- Profit Maximization
- Supply
- Market Eq.

ECON 101
Sp 2023



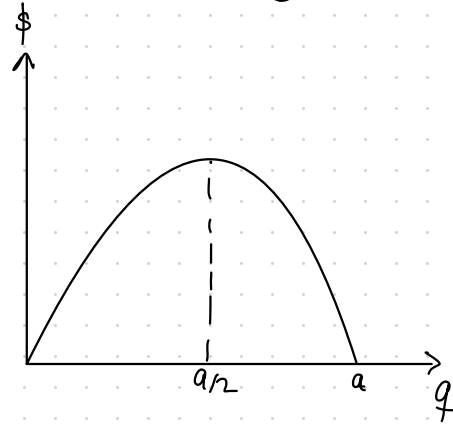
Revenues:

Given a market demand curve that a firm faces $D(p) \leftrightarrow p = D^{-1}(q)$

$$\begin{aligned} \text{Revenues}(q) &= \text{price} \times \text{quantity} \\ &= D^{-1}(q) \times q \end{aligned}$$

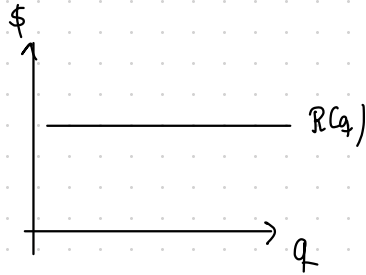
For a linear demand curve $q = a - bp \leftrightarrow p = \frac{a}{b} - \frac{q}{b}$

$$R(q) = \left(\frac{a}{b} - \frac{q}{b} \right) q = \frac{a}{b} q - \frac{q^2}{b}$$

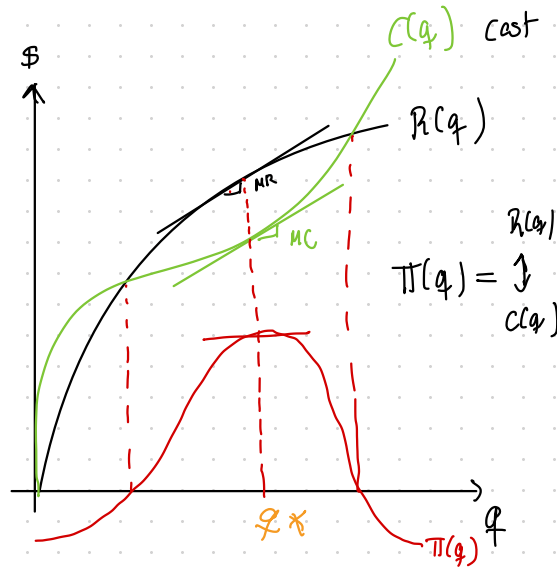


$$q = \frac{1}{P} \Rightarrow P = \frac{1}{q}$$

$$R(q) = \left(\frac{1}{q}\right) \cdot q = 1$$



$$\text{Profits: } \Pi(q) = R(q) - C(q)$$



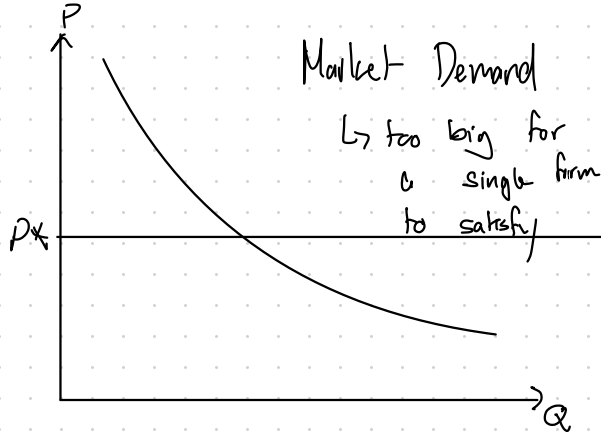
cost function.

$$\begin{aligned} MR(q) &= \text{slope of } R(q) \\ &= \frac{dR(q)}{dq} \end{aligned}$$

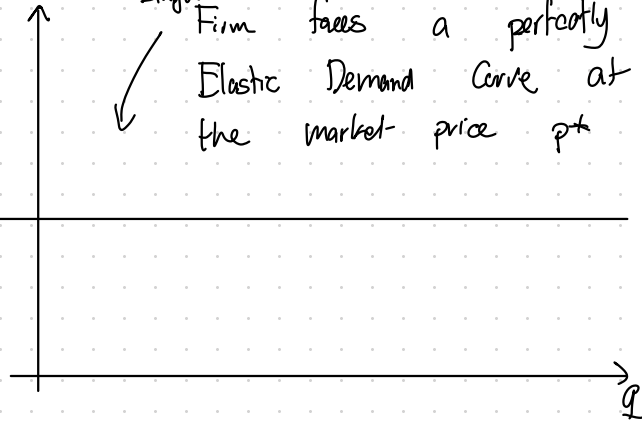
$$\begin{aligned} MC(q) &= \text{slope of } C(q) \\ &= \frac{dC(q)}{dq} \end{aligned}$$

$$\text{Optimality Condition: } MR(q^*) = MC(q^*)$$

The Demand Curve faced by a Competitive Firm

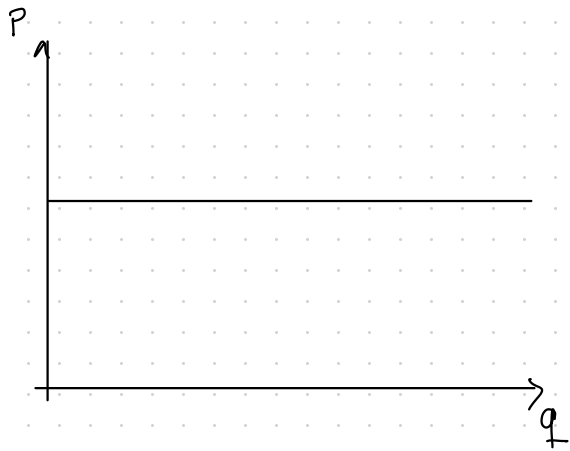


Single Firm faces a perfectly Elastic Demand Curve at the market price P_k



Perfect Competition

- No barriers to entry
- Homogeneous product
- Firm small compared to whole market.



$$MR = AR = P$$

$$MR(q) = P$$

$$AR(q) = \frac{R(q)}{q} = \frac{P \cdot q}{q} = P$$

$$R(q) = P \cdot q$$

↳ A perfectly competitive firm takes the market price as given.

$$\max_q P \cdot q - C(q)$$

Suppose we are analyzing the Short-Run, $C(q)$ are Short-Run costs.

Optimality condition: $MR = MC$

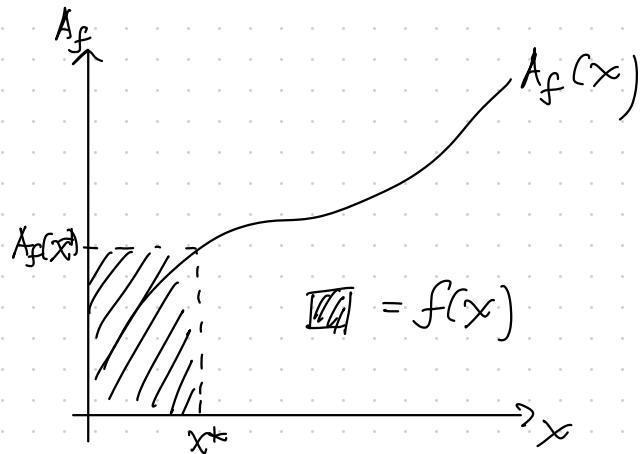
$$P = MC(q)$$

This is the (inverse) supply curve of the firm!

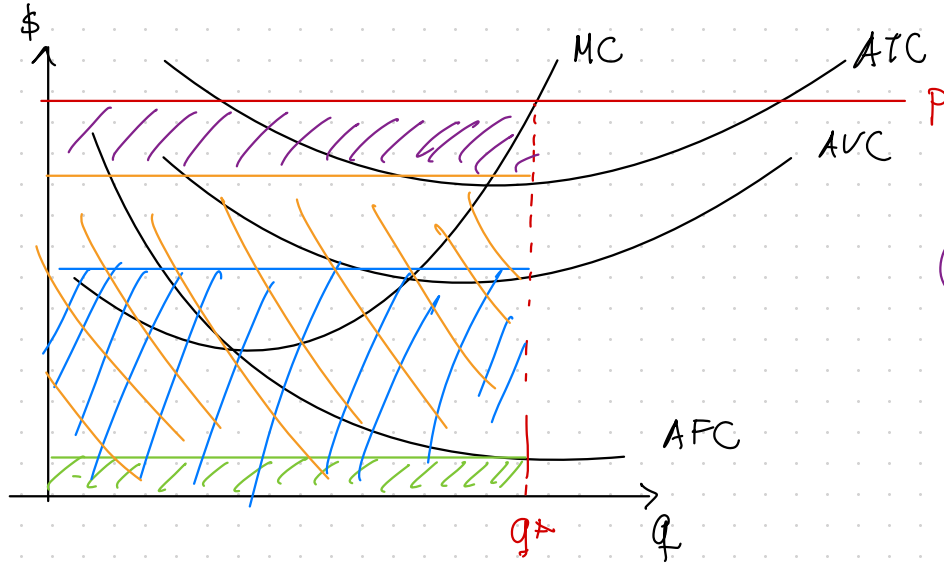
Small- Aside.

$$f(x) \quad A_f(x) = \frac{f(x)}{x}$$

If we multiply x . $A_f(x) = f(x)$



Consider the following cost structure:

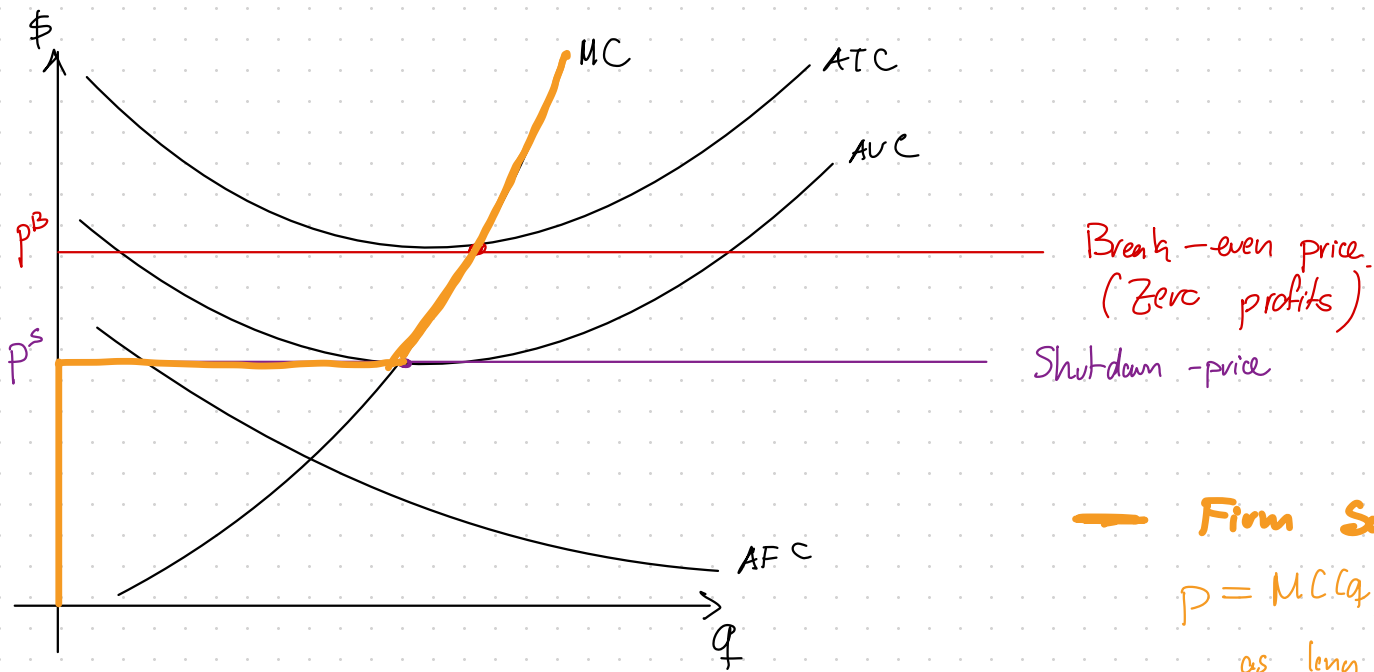


 FC

 VC

 TC

 + all above R
 → Profits



Firm Supply
 $P = MC(q)$
 as long $P \geq p^S$

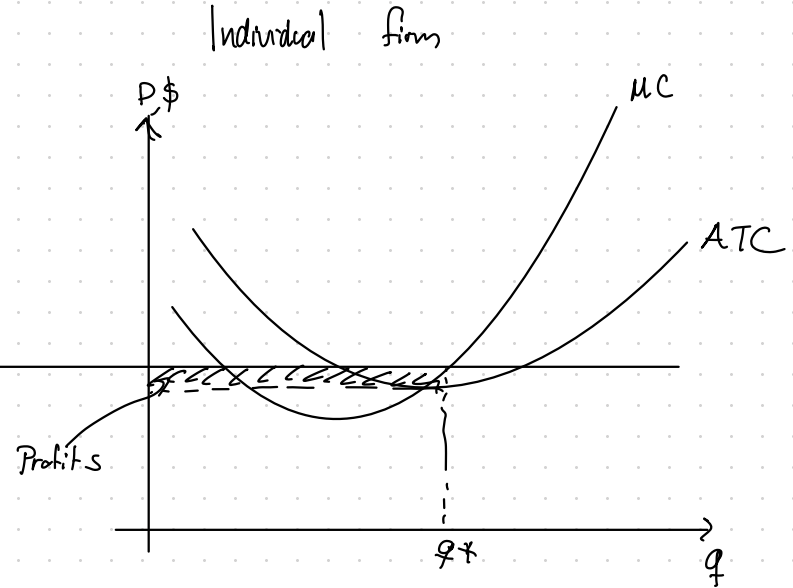
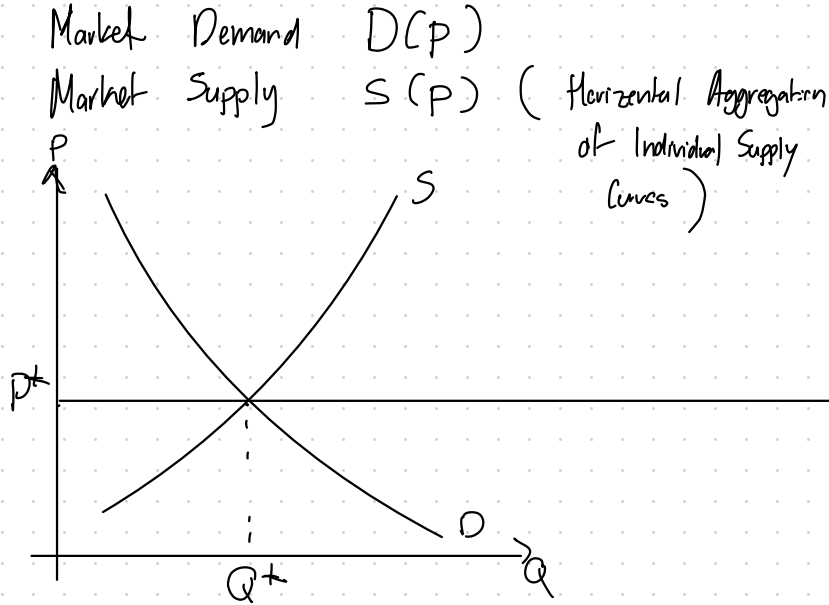
$P < p^S \rightarrow$ The firm doesn't operate since it is not able to cover its VC

$P > p^B \rightarrow$ Firm makes positive profits

$p^S < P < p^B \rightarrow$ Firm makes losses, but can cover its VC
 $\rightarrow$ decides to operate

Market Equilibrium

Assume all firms are identical



$$Q^* = \sum_{i=1}^N q^*$$

Long-Run Equilibrium : $p^{*}_{LR} = p^{BE} \rightarrow$ No Entry / No exit.

The long-run eq. condition is making zero profits

